

Physical Chemistry (I)

Lecture 21

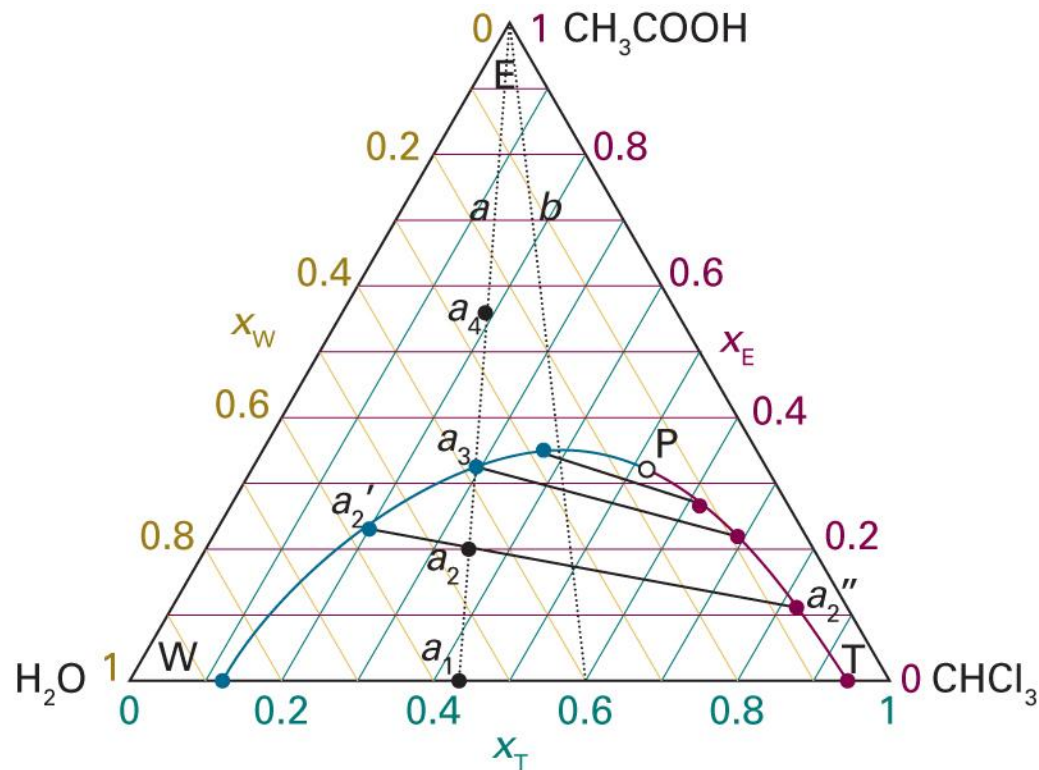


In the Last Class

- Phase diagrams of ternary systems
- Chemical reactions and equilibria
 - Reaction quotient and equilibrium constant
 - Activities
- Le Chatelier's principle
 - Amount change
 - Volume change
 - Temperature change



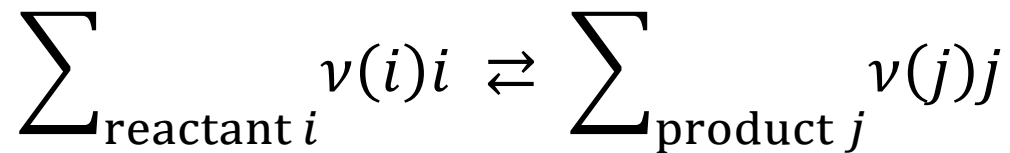
Partially Miscible Liquids



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5E.4)

Chemical Equilibrium

For an arbitrary reaction,



Starting from

$$dG = \sum_{\text{product } j} \mu_j dn_j + \sum_{\text{reactant } i} \mu_i dn_i$$

we obtain

$$\Delta_r G = \sum_{\text{product } j} \mu_j \nu_j - \sum_{\text{reactant } i} \mu_i \nu_i$$

Chemical Potential and Activity

$$\mu = \mu^\ominus + RT \ln a$$

Component	Basis	Standard state	Activity	Limits
Solid or liquid		Pure, 1 bar	$a = 1$	
Solvent	Raoult	Pure solvent, 1 bar	$a = p/p^*, a = \gamma x$	$\gamma \rightarrow 1$ as $x \rightarrow 1$ (pure solvent)
Solute	Henry	(1) A hypothetical state of the pure solute	$a = p/K, a = \gamma x$	$\gamma \rightarrow 1$ as $x \rightarrow 0$
		(2) A hypothetical state of the solute at molality $b^\ominus$	$a = \gamma b/b^\ominus$	$\gamma \rightarrow 1$ as $b \rightarrow 0$
Gas	Fugacity [†]	Pure, a hypothetical state of 1 bar and behaving as a perfect gas	$f = \gamma p$	$\gamma \rightarrow 1$ as $p \rightarrow 0$

$$\Delta_r G = \Delta_r G^\ominus + RT \ln \left(\prod_{\text{product } j} a_j^{\nu_j} / \prod_{\text{reactant } i} a_i^{\nu_i} \right)$$



Q and K

$$\Delta_r G = \Delta_r G^\ominus + RT \ln Q$$

where the **reaction quotient** Q is defined as

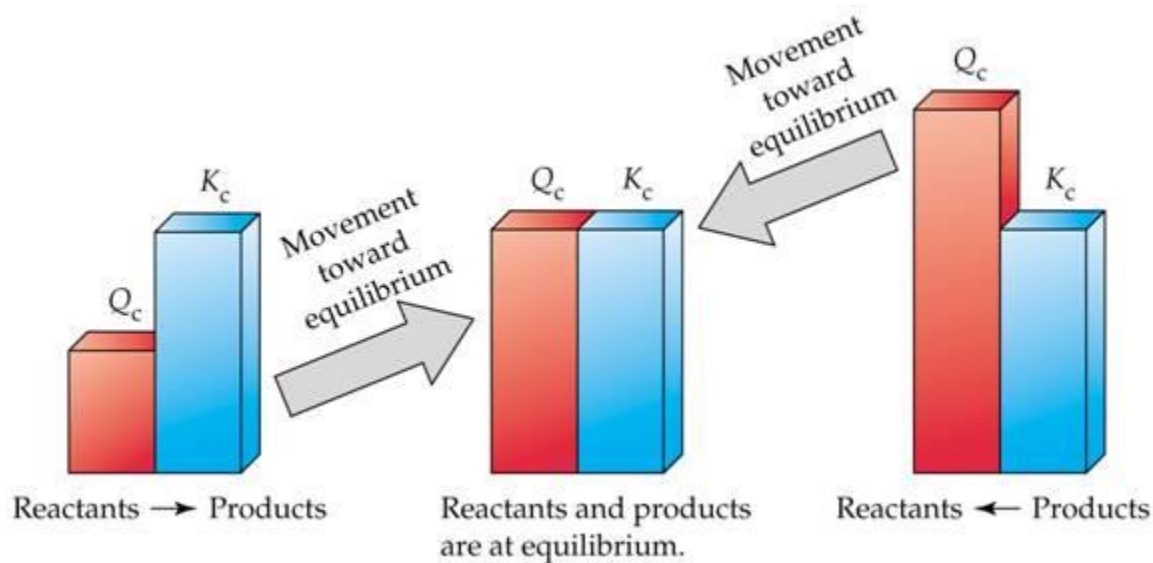
$$Q = \frac{\prod_{\text{product } j} a_j^{v_j}}{\prod_{\text{reactant } i} a_i^{v_i}}$$

At equilibrium, $\Delta_r G = 0$

$$0 = \Delta_r G^\ominus + RT \ln Q_{\text{eq}} = \Delta_r G^\ominus + RT \ln K$$

where the **equilibrium constant** K is Q at equilibrium

Meaning of Q and K



Le Chatelier's Principle

When subjected to a disturbance,
a system at equilibrium tends to respond in a way
that **minimizes the effect of the disturbance**.

Disturbances to consider:

- Amount (concentration, pressure)
- Temperature

Derivations

- Q and K : direction of reaction
- Amount change: change of Q
- Volume change: change of Q
- Temperature change: change of K
 - van't Hoff equation

$$\frac{d \ln K}{dT} = \frac{\Delta_r H^\ominus}{RT^2}$$

9. Electrochemistry



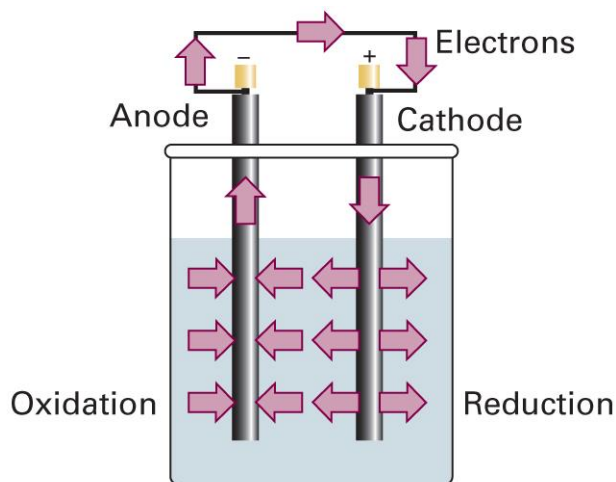
Chapter Overview

- Electrochemical cells
 - Basic terminology and concepts
 - Thermodynamics of cells
 - Electrode potentials
- Physical chemistry of ions
 - Gibbs energy of solvation
 - Screening effects
 - Activities of ions



Electrochemical Cell

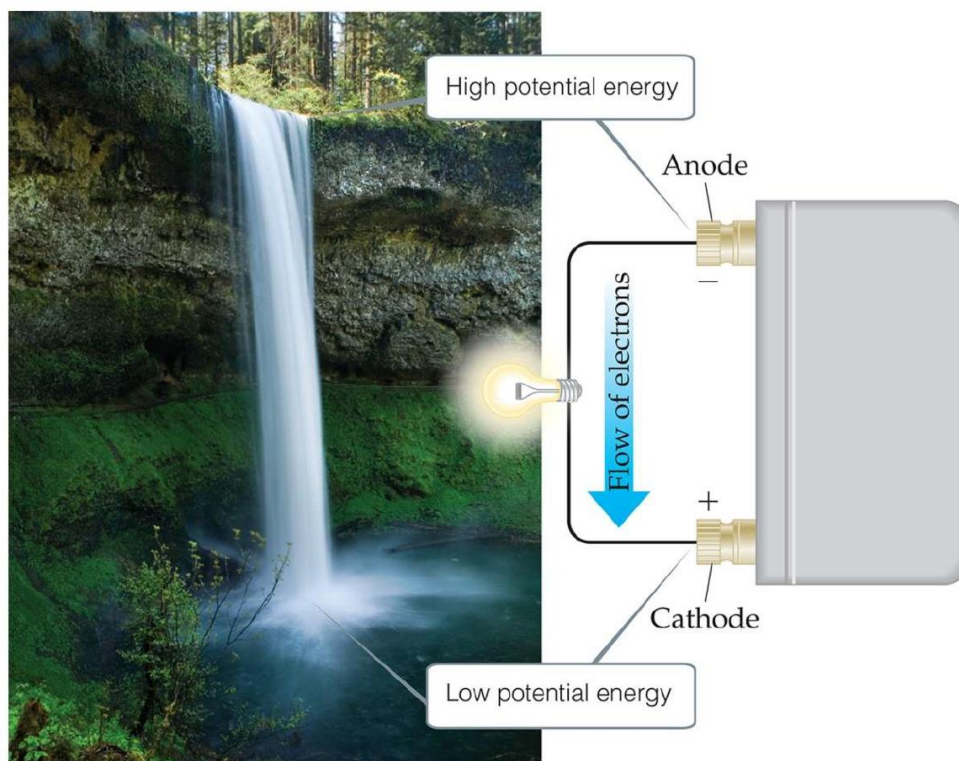
- Two electrodes + electrolytes
 - Anode: oxidation
 - Cathode: reduction



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 6C.1)

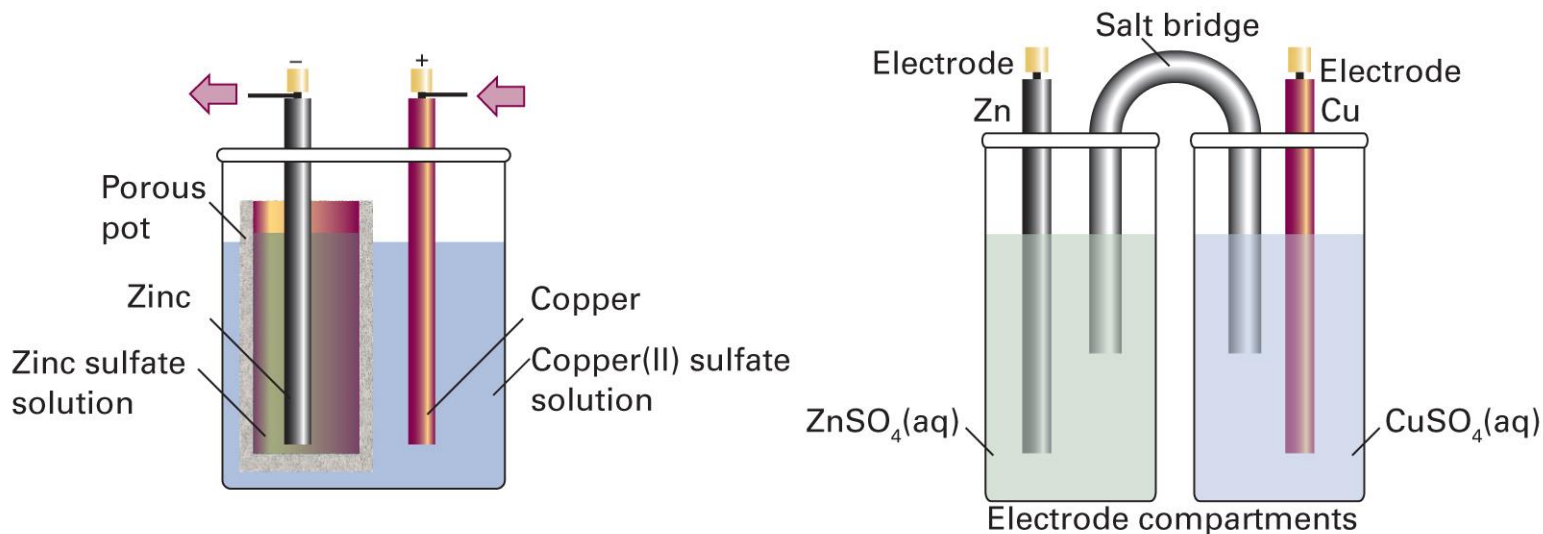
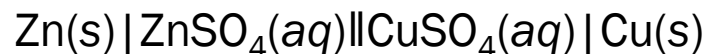
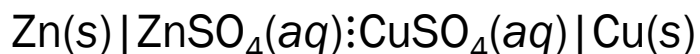
Cell Potential

Potential difference between the two electrodes



(Image: <https://www.unf.edu/~michael.lufaso/chem2046/2046chapter20.pdf>)

Liquid Junction and Salt Bridge



Liquid junction may have an **additional potential difference**

Thermodynamics of Cell

Recall that

$$dG_{T,p} = \delta w_{\text{add,max}} = \delta w_{\text{elec}}$$

for a **reversible** electrochemical reaction.

For an infinitesimal progress of the reaction $d\xi$,

$$dG = \Delta_r G d\xi$$

Hence,

$$\delta w_{\text{elec}} = \Delta_r G d\xi$$

Thermodynamics of Cell

$$\delta w_{\text{elec}} = \Delta_r G d\xi$$

On the other hand, the electrical work is

$$\delta w_{\text{elec}} = \phi dQ$$

where ϕ is the electric field and dQ is the transferred charge. For the electrochemical reaction,

$$\phi = E_{\text{cell}}, \quad dQ = -\nu e N_A d\xi = -\nu F d\xi$$

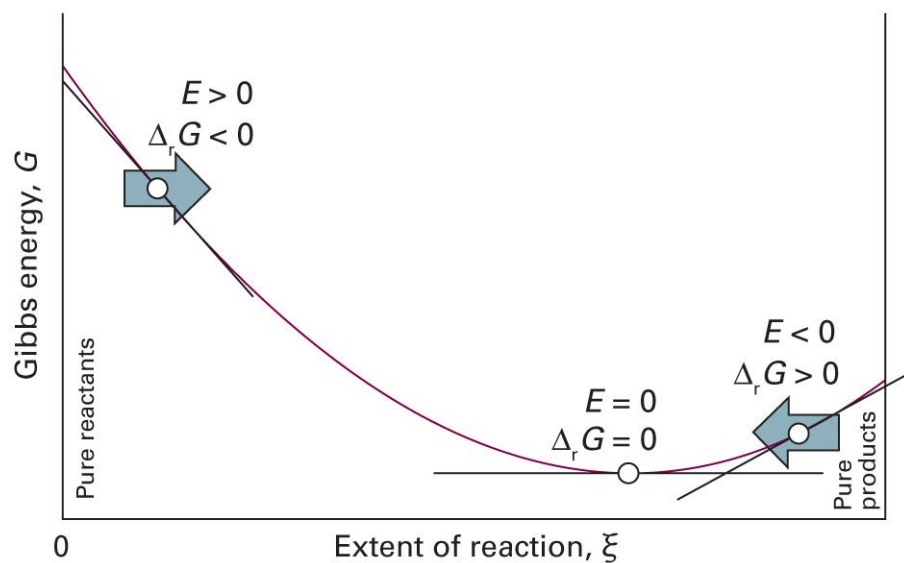
$$\delta w_{\text{elec}} = -\nu F E_{\text{cell}} d\xi$$

Thus, $\Delta_r G = -\nu F E_{\text{cell}}$

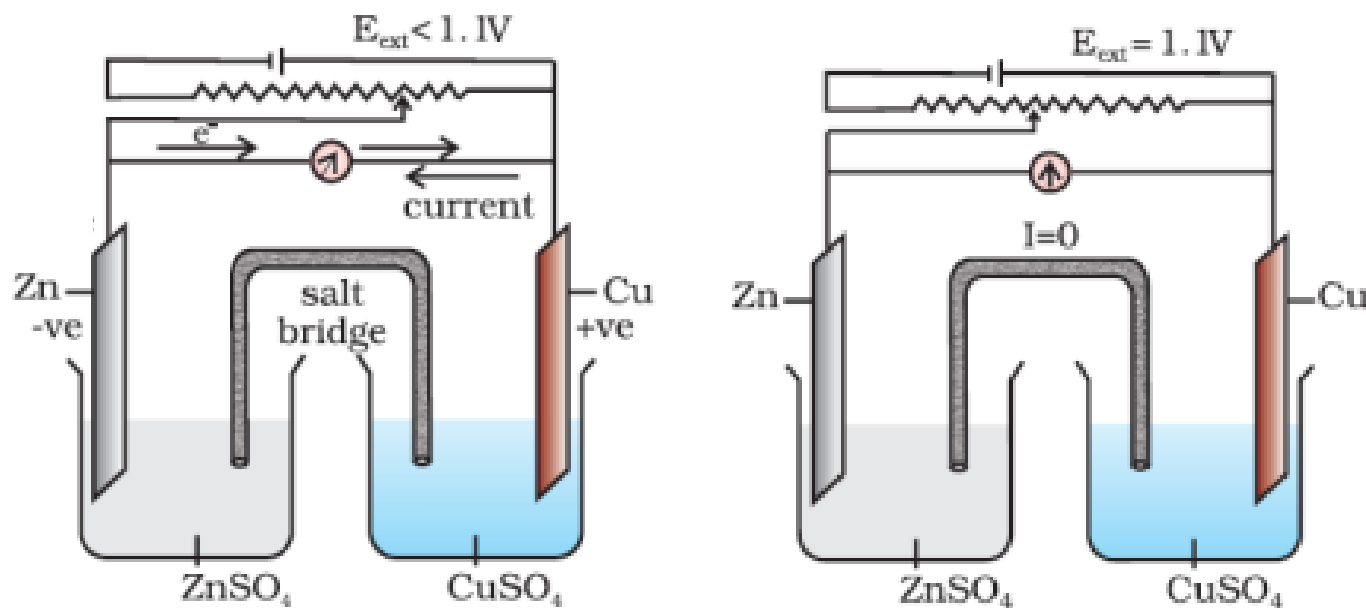


Cell Potential and Reaction

$$\Delta_r G = -\nu F E_{\text{cell}}$$



Measurement of Cell Potential



$$E_{\text{cell}} = (E_{\text{ext}})_{I=0}$$

Nernst Equation

$$\Delta_r G = -\nu F E_{\text{cell}}$$

Using $\Delta_r G = \Delta_r G^\ominus + RT \ln Q$,

$$-\nu F E_{\text{cell}} = \Delta_r G^\ominus + RT \ln Q$$

Thus,

$$E_{\text{cell}} = E_{\text{cell}}^\ominus - \frac{RT}{\nu F} \ln Q$$

where $E_{\text{cell}}^\ominus = -\Delta_r G^\ominus / \nu F$

At Equilibrium

$$E_{\text{cell}} = E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln Q$$

At equilibrium, $E_{\text{cell}} = 0$ and $Q = K$

$$0 = E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln K$$

or,

$$E_{\text{cell}}^{\ominus} = \frac{RT}{\nu F} \ln K$$

Brief Illustration 6C.8

Calculate the standard reaction Gibbs energy of the reaction $\text{Ag}^+(\text{aq}) + (1/2)\text{H}_2(\text{g}) \rightarrow \text{H}^+(\text{aq}) + \text{Ag}(\text{s})$, using the standard cell potential $E_{\text{cell}}^\ominus = +0.7996 \text{ V}$ of the cell $\text{Pt}(\text{s}) | \text{H}_2(\text{g}) | \text{H}^+(\text{aq}) || \text{Ag}^+(\text{aq}) | \text{Ag}(\text{s})$.

Since $\Delta_{\text{r}}G = -\nu F E_{\text{cell}}$,

$$\Delta_{\text{r}}G^\ominus = -\nu F E_{\text{cell}}^\ominus$$

$$= -1 \times (9.6485 \times 10^4 \text{ C mol}^{-1}) \times (0.7996 \text{ V}) = 77.15 \text{ kJ mol}^{-1}$$

Temperature Coefficient

$$E_{\text{cell}}^{\ominus} = -\frac{\Delta_{\text{r}}G^{\ominus}}{\nu F}$$

Since $(\partial G / \partial T)_p = -S$,

$$\frac{dE_{\text{cell}}^{\ominus}}{dT} = -\frac{1}{\nu F} \left(\frac{\partial \Delta_{\text{r}}G^{\ominus}}{\partial T} \right)_p = \frac{\Delta_{\text{r}}S^{\ominus}}{\nu F}$$

Furthermore,

$$\Delta_{\text{r}}H^{\ominus} = \Delta_{\text{r}}G^{\ominus} + T\Delta_{\text{r}}S^{\ominus} = -\nu F \left(E_{\text{cell}}^{\ominus} - T \frac{dE_{\text{cell}}^{\ominus}}{dT} \right)$$

Example 6C.1

The standard potential of the cell $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{HBr(aq)} | \text{AgBr(s)} | \text{Ag(s)}$ was measured over a range of temperatures, and the data were found to fit the following polynomial:

$$E_{\text{cell}}^{\ominus}/\text{V} = 0.07131 - 4.99 \times 10^{-4}(T/\text{K} - 298) - 3.45 \times 10^{-6}(T/\text{K} - 298)^2$$

The cell reaction is $\text{AgBr(s)} + (1/2)\text{H}_2(\text{g}) \rightarrow \text{Ag(s)} + \text{HBr(aq)}$, and has $\nu = 1$. Evaluate the standard reaction Gibbs energy, enthalpy, and entropy at 298 K.

At 298 K, $E_{\text{cell}}^{\ominus} = 0.07131 \text{ V}$ and

$$\Delta_{\text{r}}G^{\ominus} = -\nu F E_{\text{cell}}^{\ominus}$$

$$= -1 \times (9.6485 \times 10^4 \text{ C mol}^{-1}) \times (0.07131 \text{ V}) = -6.880 \text{ kJ mol}^{-1}$$

Example 6C.1

On the other hand,

$$dE_{\text{cell}}^{\ominus}/dT = -4.99 \times 10^{-4} \text{ V K}^{-1} - 2 \times 3.45 \times 10^{-6}(T/\text{K} - 298) \text{ V K}^{-1}$$

At 298 K, $dE_{\text{cell}}^{\ominus}/dT = -4.99 \times 10^{-4} \text{ V K}^{-1}$ and

$$\Delta_{\text{r}}S^{\ominus} = \nu F(dE_{\text{cell}}^{\ominus}/dT)$$

$$= 1 \times (9.6485 \times 10^4 \text{ C mol}^{-1}) \times (-4.99 \times 10^{-4} \text{ V K}^{-1}) = -48.1 \text{ J K}^{-1} \text{ mol}^{-1}$$

Finally,

$$\Delta_{\text{r}}H^{\ominus} = \Delta_{\text{r}}G^{\ominus} + T\Delta_{\text{r}}S^{\ominus}$$

$$= (-6.880 \text{ kJ mol}^{-1}) + (298 \text{ K})(-48.1 \text{ J K}^{-1} \text{ mol}^{-1}) = -21.2 \text{ kJ mol}^{-1}$$

Electrode Potentials

Cell potential = cathode contribution + anode contribution

For a cell L||R,

$$E_{\text{cell}}^{\ominus} = E^{\ominus}(\text{R}) - E^{\ominus}(\text{L})$$

Impossible to measure each contribution

Convention

- Standard hydrogen electrode (SHE): $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{H}^+(\text{aq})$

$E^\ominus = 0$ at all temperatures

- Standard potential $E^\ominus(\text{X})$

For a cell $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{H}^+(\text{aq}) || \text{X}$, $E^\ominus(\text{X}) = E_{\text{cell}}^\ominus$

Standard Potentials

Couple	$E^\ominus/\text{V}$
$\text{Ce}^{4+}(\text{aq}) + \text{e}^- \rightarrow \text{Ce}^{3+}(\text{aq})$	+1.61
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$	+0.34
$\text{AgCl}(\text{s}) + \text{e}^- \rightarrow \text{Ag}(\text{s}) + \text{Cl}^-(\text{aq})$	+0.22
$\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \frac{1}{2}\text{H}_2(\text{g})$	0
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$	-0.76
$\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}(\text{s})$	-2.71

Thermodynamics of Potentials

Each electrode potential can be converted into

$$-vFE^{\ominus} = \Delta_r G^{\ominus}$$

When a combination of reactions a and b gives reaction c,

$$\Delta_r G^{\ominus}(c) = \Delta_r G^{\ominus}(a) + \Delta_r G^{\ominus}(b)$$

$$-v_c F E^{\ominus}(c) = -v_a F E^{\ominus}(a) - v_b F E^{\ominus}(b)$$

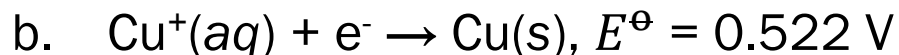
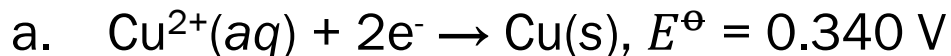
Thus,

$$v_c E^{\ominus}(c) = v_a E^{\ominus}(a) + v_b E^{\ominus}(b)$$

Example 6D.2

Given that the standard potentials of the Cu^{2+}/Cu and Cu^+/Cu couples are +0.340 V and +0.522 V, respectively, evaluate $E^\ominus(\text{Cu}^{2+}, \text{Cu}^+)$.

The half-reactions are



The target reaction $\text{Cu}^{2+}(\text{aq}) + \text{e}^- \rightarrow \text{Cu}^+(\text{aq})$ can be obtained by a – b:

$$\nu_{\text{c}} E^\ominus(\text{c}) = \nu_{\text{a}} E^\ominus(\text{a}) - \nu_{\text{b}} E^\ominus(\text{b})$$

$$E^\ominus(\text{c}) = 2E^\ominus(\text{a}) - E^\ominus(\text{b}) = 2 \times (0.340 \text{ V}) - (0.522 \text{ V}) = 0.158 \text{ V}$$

Reducing Power

For two redox couples $\text{Ox}_\text{L}/\text{Red}_\text{L}$ and $\text{Ox}_\text{R}/\text{Red}_\text{R}$,
the cell reaction $\text{Red}_\text{L} + \text{Ox}_\text{R} \rightarrow \text{Ox}_\text{L} + \text{Red}_\text{R}$ has

- $K > 1$: $E^\ominus_{\text{cell}} > 0$ and $E^\ominus(\text{L}) < E^\ominus(\text{R})$
- $K < 1$: $E^\ominus_{\text{cell}} < 0$ and $E^\ominus(\text{L}) > E^\ominus(\text{R})$

Red_L has a thermodynamic tendency to reduce Ox_R
if $E^\ominus(\text{L}) < E^\ominus(\text{R})$

Electrochemical Series

Least strongly reducing

Gold (Au^{3+}/Au)

Platinum (Pt^{2+}/Pt)

Silver (Ag^+/Ag)

Mercury (Hg^{2+}/Hg)

Copper (Cu^{2+}/Cu)

Hydrogen (H^+/H_2)

Tin (Sn^{2+}/Sn)

Nickel (Ni^{2+}/Ni)

Iron (Fe^{2+}/Fe)

Zinc (Zn^{2+}/Zn)

Chromium (Cr^{3+}/Cr)

Aluminium (Al^{3+}/Al)

Magnesium (Mg^{2+}/Mg)

Sodium (Na^+/Na)

Calcium (Ca^{2+}/Ca)

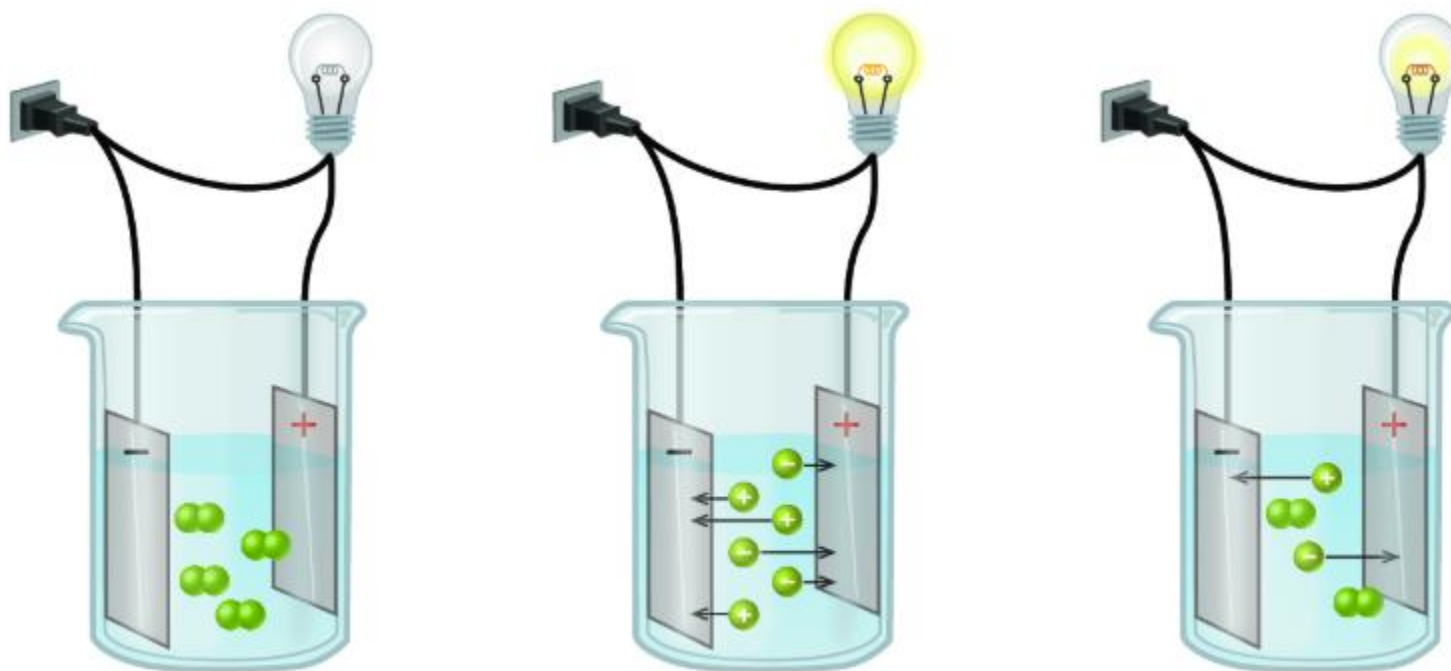
Potassium (K^+/K)

Most strongly reducing



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Cells and Electrolytes



Physical Chemistry of Ions

Easy to model: charge-charge interactions

For two electric point charges Q_1 and Q_2 separated by r , the potential energy is given as

$$E_p(r) = \frac{Q_1 Q_2}{4\pi\epsilon r}$$

where ϵ is the permittivity of the medium.

Coulomb Potential

$$E_p(r) = \frac{Q_1 Q_2}{4\pi\epsilon r}$$

For the point charge Q_2 , the potential due to Q_1 is

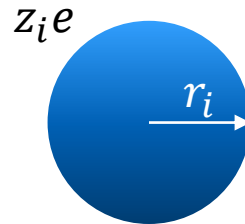
$$\phi(r) = \frac{Q_1}{4\pi\epsilon r}$$

and

$$E_p(r) = Q_2 \phi(r)$$

Work for Charging an Ion

Model each ion i as a sphere with radius r_i and charge $z_i e$.



The electric work required to charge this ion is

$$\delta w_{\text{elec}} = \phi dQ$$

$$w_{\text{elec}} = \int_0^{z_i e} \phi dQ = \int_0^{z_i e} \left(\frac{Q}{4\pi\epsilon r_i} \right) dQ = \frac{z_i^2 e^2}{8\pi\epsilon r_i}$$

Gibbs Energy for Solvation

$$w_{\text{elec}} = \frac{z_i^2 e^2}{8\pi\epsilon r_i}$$

$$\Delta_{\text{solv}} G^\ominus = w(\text{medium}) - w(\text{vacuum})$$

For vacuum, $\epsilon = \epsilon_0$ and

$$\Delta_{\text{solv}} G^\ominus = \frac{z_i^2 e^2}{8\pi\epsilon r_i} - \frac{z_i^2 e^2}{8\pi\epsilon_0 r_i} = -\frac{z_i^2 e^2}{8\pi\epsilon_0 r_i} \left(1 - \frac{\epsilon_0}{\epsilon}\right)$$

Born Equation

$$\Delta_{\text{solv}} G^{\ominus} = -\frac{z_i^2 e^2}{8\pi\epsilon_0 r_i} \left(1 - \frac{1}{\epsilon_r}\right)$$

where the relative permittivity $\epsilon_r = \epsilon/\epsilon_0$.

- $\Delta_{\text{solv}} G^{\ominus} < 0$: solvation of ions is favored
- Water has $\epsilon_r = 78.54$ at 25°C, and

$$\Delta_{\text{solv}} G^{\ominus} = -\frac{z_i^2}{r_i/\text{pm}} \times (6.86 \times 10^4 \text{ kJ mol}^{-1})$$

Brief Illustration 3D.3

Estimate the difference in the values of $\Delta_{\text{solv}}G^\ominus$ for Cl^- and I^- in water at 25°C , given their radii as 181 pm and 220 pm, respectively.

$$\Delta_{\text{solv}}G^\ominus = -\frac{z_i^2}{r_i/\text{pm}} \times (6.86 \times 10^4 \text{ kJ mol}^{-1})$$

Thus,

$$\begin{aligned}\Delta_{\text{solv}}G^\ominus(\text{Cl}^-) - \Delta_{\text{solv}}G^\ominus(\text{I}^-) &= -\left(\frac{1}{181} - \frac{1}{220}\right) \times (6.86 \times 10^4 \text{ kJ mol}^{-1}) \\ &= -67 \text{ kJ mol}^{-1}\end{aligned}$$

Generalized Born Model

$$\Delta_{\text{solv}} G^\ominus = -\frac{z_i^2 e^2}{8\pi\epsilon_0} \left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{r_i}$$

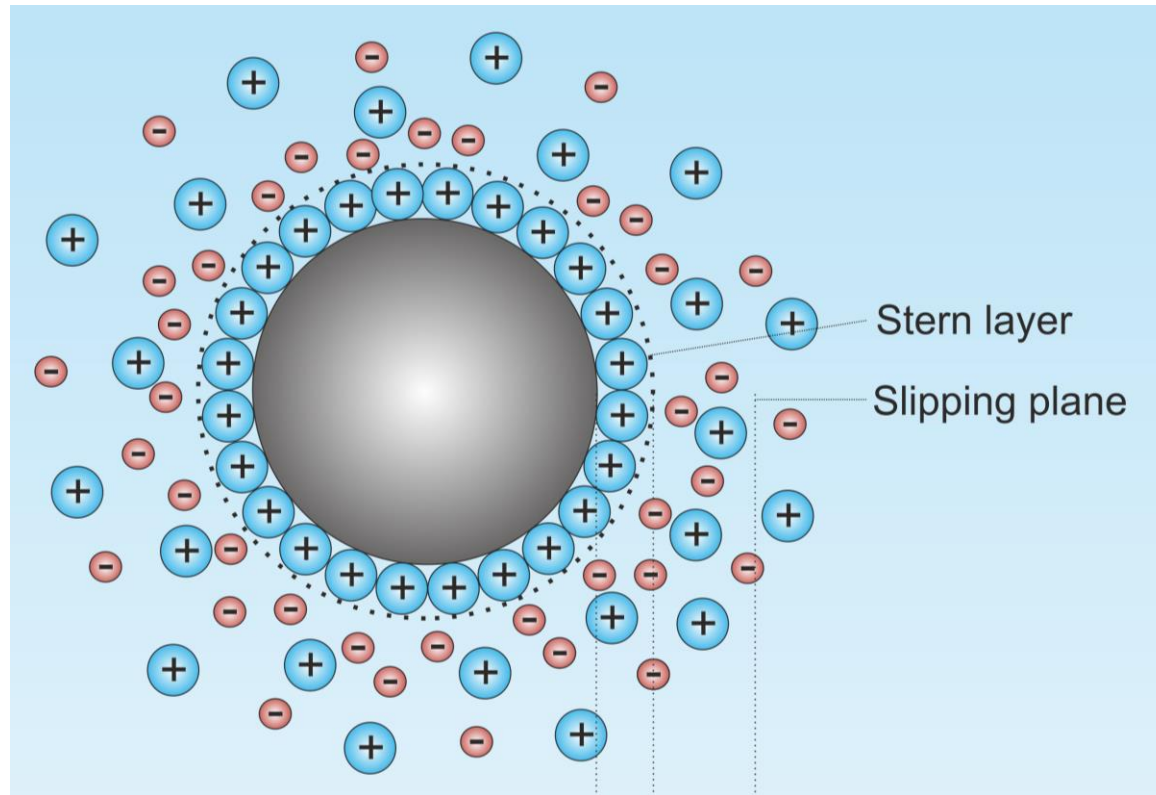
We can generalize the Born model by using

$$\Delta_{\text{solv}} G^\ominus = -\frac{z_i^2 e^2}{8\pi\epsilon_0} \left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{\sqrt{r_i^2 + a_i^2 e^{-(r_i/2a_i)^2}}}$$

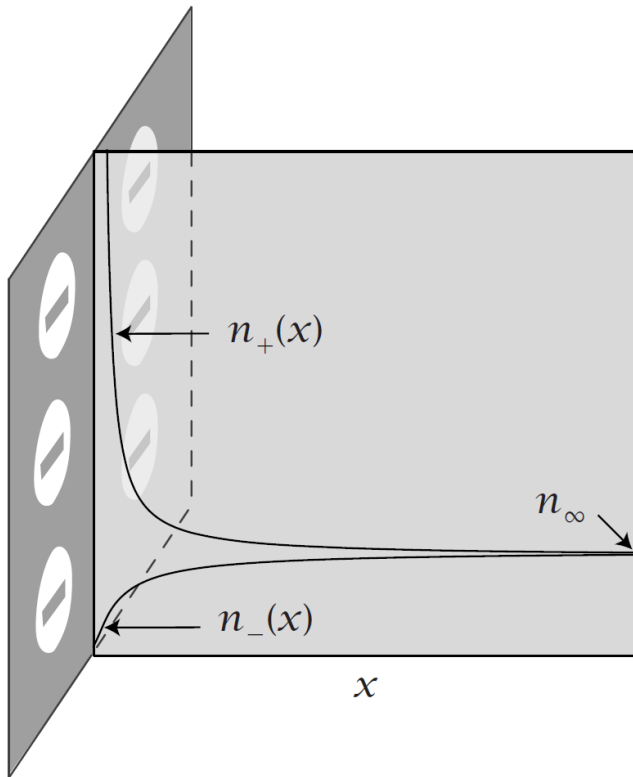
where a_i is the effective Born radius.

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Screening Effects of Ions



Boltzmann Distribution



For an ion of charge Q ,

- Potential energy

$$E_p = Q\phi$$

- Boltzmann distribution

$$P \propto e^{-E_p/k_B T} = e^{-Q\phi/k_B T}$$

$$n_+ = n_\infty e^{-ze\phi/k_B T}$$

$$n_- = n_\infty e^{+ze\phi/k_B T}$$

Poisson-Boltzmann Equation

- Charge density

$$\begin{aligned}\rho &= zeN_A n_+/V + (-ze)N_A n_-/V = zF(n_+ - n_-)/V \\ &= zFn_\infty(e^{-ze\phi/k_BT} - e^{+ze\phi/k_BT})/V\end{aligned}$$

- Poisson equation: relationship between $\rho(x)$ and $\phi(x)$

$$\begin{aligned}\nabla^2 \phi &= -\rho/\varepsilon \\ (\nabla^2 &= \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2)\end{aligned}$$

- Poisson-Boltzmann equation

$$\nabla^2 \phi = \frac{zFn_\infty}{\varepsilon V} (e^{+ze\phi/k_BT} - e^{-ze\phi/k_BT})$$

Debye-Hückel Equation

$$\nabla^2 \phi = \frac{zF n_{\infty}}{\epsilon V} (e^{+ze\phi/k_B T} - e^{-ze\phi/k_B T})$$

At the limit of a dilute solution, $ze\phi/k_B T \ll 1$ and

$$e^{\pm ze\phi/k_B T} \approx 1 \pm ze\phi/k_B T = 1 \pm zF\phi/RT$$

Hence,

$$\nabla^2 \phi = \frac{zF n_{\infty}}{\epsilon V} \{(1 + zF\phi/RT) - (1 - zF\phi/RT)\}$$

$$\nabla^2 \phi = \frac{2z^2 F^2 n_{\infty}}{\epsilon V R T} \phi$$

Ion Strength

$$I = \frac{1}{2} \sum_i z_i^2 \left(\frac{b_i}{b^\ominus} \right)$$

For two types of ions with $z_+ = z_- = z$,

$$I = \frac{z_+^2 b_+ + z_-^2 b_-}{2b^\ominus} = \frac{z^2 b}{b^\ominus}$$

Since $b = n_B/m_A = n_\infty/m_A$,

$$I = \frac{z^2 n_\infty}{b^\ominus m_A}$$

Generalization of D-H Eq.

$$\nabla^2 \phi = \frac{2F^2 z^2 n_{\infty}}{\varepsilon V R T} \phi$$

Since the ion strength is $I = z^2 n_{\infty} / b^{\ominus} m_A$,

$$\nabla^2 \phi = \frac{2F^2 I b^{\ominus} (m_A / V)}{\varepsilon R T} \phi$$

This equation is generally applicable to any salt system.

Debye Length

$$r_D = \sqrt{\frac{\varepsilon RT}{2F^2 I b^\ominus(m_A/V)}} \propto \frac{1}{\sqrt{I}}$$

Then

$$\nabla^2 \phi = \phi / r_D^2$$

For an aqueous solution at room temperature,

$$r_D \approx (0.304 \text{ nm M}^{1/2}) / \sqrt{I}$$

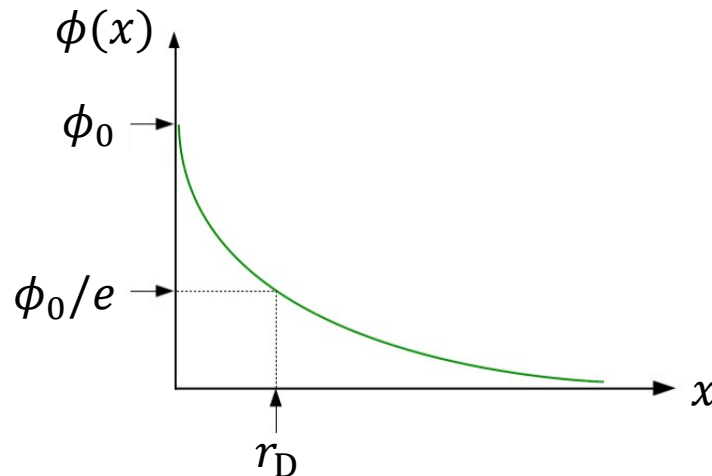
Ions Around Electrodes

For a charged plate, solving the differential equation

$$\nabla^2 \phi = \phi / r_D^2$$

gives a solution of

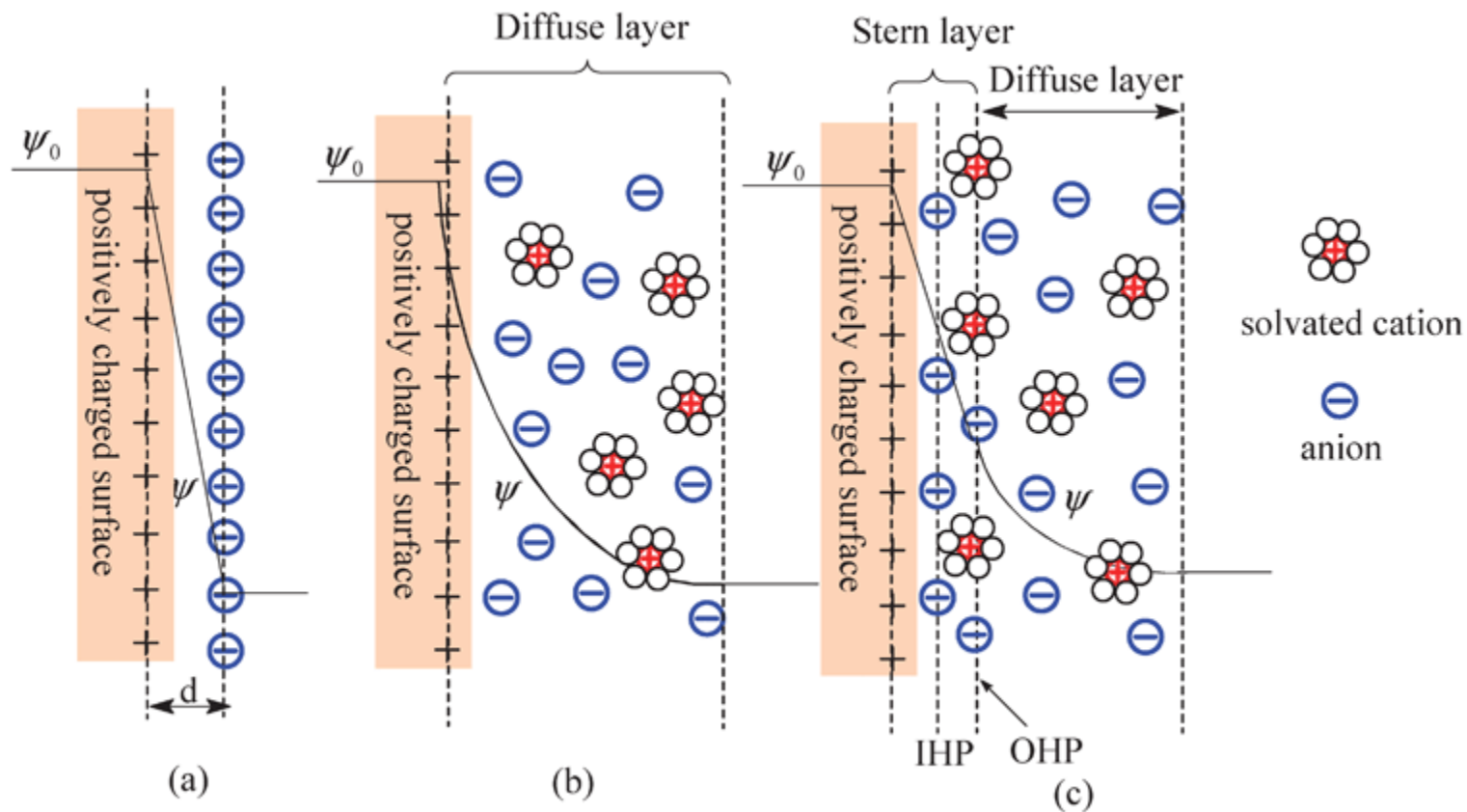
$$\phi(x) = \phi_0 e^{-x/r_D}$$



(Image: <https://docplayer.net/21695464-Charged-interfaces-formation-of-charged-surfaces-helmholz-gouy-chapman-and-stern-model-of-double-layer.html>)

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Ions Around Electrodes



(Image: Zhang and Zhao, *Chemical Society Reviews* 38: 2520-2531 (2009), Figure 2)

Screening Effect

For a point charge of Q , solving the differential equation

$$\nabla^2 \phi = \phi / r_D^2$$

gives a solution of

$$\phi(r) = \frac{Q}{4\pi\epsilon r} e^{-r/r_D}$$

which can be compared to the Coulombic potential

$$\phi(r) = \frac{Q}{4\pi\epsilon r}$$

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Screening Effect

